

Education

08/2018 – 11/2023	Ph.D. in Computer Architecture , Chalmers University of Technology (Sweden) Ph.D. Thesis: <i>Adaptive Task Scheduling and Resource Management Techniques for Energy Efficiency in Multi-core Systems</i>
09/2015 – 07/2018	M.S. in Computer Engineering , National University of Defense Technology (China) M.S. Thesis: <i>Implementing and Optimizing Alternating Least Squares on Many-Cores</i>
09/2011 – 06/2015	B.S. in Computer Science and Technology , Southwest Jiaotong University (China) B.S. Thesis: <i>Design of Interest Mining System Based on WEB Text</i>

Work experience

05/2024 – 05/2027	Postdoctoral researcher , Chalmers University of Technology (Sweden)
12/2023 – 04/2024	Research Assistant , Chalmers University of Technology (Sweden)

Projects

08/2025 –	Energy Efficient LLM Disaggregation on Heterogeneous Multi-GPUs Chalmers University of Technology (Sweden) <i>Prefill-decode Disaggregation, vLLM Serving, Request Routing, DVFS, Tensor Parallelism, Instance Scaling, KV-cache management</i>
04/2025 –	Joint Task Placement and DVFS for Energy-Efficient Multi-GPU Runtimes Chalmers University of Technology (Sweden) <i>GPU Performance and Power Modeling, NVML kernel Profiling, CUDA Tasking model, Scheduling Scheme Optimization for Performance and Energy Efficiency</i>

03/2025 –	DARE: A new era for supercomputing in Europe Chalmers University of Technology (Sweden) <i>SYCL Compiler Development (AdaptiveCpp, OpenMP Backend)</i>
12/2021 –	EUPilot: Pilot using Independent Local & Open Technologies Chalmers University of Technology (Sweden) <i>High-performance OpenMP Task Runtime Scheduler Development in LLVM</i>
04/2021 – 07/2025	eProcessor - An Open Source Full Stack Ecosystem Chalmers University of Technology (Sweden) <i>Adaptive Task Scheduling for Energy Performance Trade-offs on Multi-core Architectures</i>
08/2018 – 12/2020	LEGaTO - The Low Energy Toolset for Heterogeneous Computing Chalmers University of Technology (Sweden) <i>Energy-Efficient Resource Management for Multi-Core Systems</i>
07/2016 – 02/2018	Performance Optimization of Alternating Least Squares (ALS) on GPUs National University of Defense Technology (China) <i>Development of Data Reuse / Reordering and Novel Compressed Sparse Matrix Format for GPU-Accelerated ALS</i>

Publications

2026	Jing Chen, Miquel Pericàs Joint Task Placement and DVFS for Energy-Efficient Multi-GPU Runtimes <i>Accepted in ACM International Conference on Supercomputing 2026 (ICS 2026)</i>
2025	Axel Carlsson, Edvin Mellberg, Jing Chen, Miquel Pericàs ILAN: The Interference- and Locality-Aware NUMA Scheduler <i>In 16th International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (SC - PMBS 2025)</i>
2024	Sonia Rani Gupta, Nikela Papadopoulou, Jing Chen, Miquel Pericàs Co-Design of Convolutional Algorithms and Long Vector RISC-V Processors for Efficient CNN Model Serving <i>Proceedings of the 53rd International Conference on Parallel Processing (ICPP 2024)</i> Jing Chen, Madhavan Manivannan, Bhavishya Goel, Miquel Pericàs SWEEP: Adaptive Task Scheduling for Exploring Energy Performance Trade-offs <i>In 38th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2024)</i>
2023	Jing Chen, Madhavan Manivannan, Bhavishya Goel, Miquel Pericàs JOSS: Joint Exploration of CPU-Memory DVFS and Task Scheduling for Energy Efficiency <i>In 52nd International Conference on Parallel Processing (ICPP 2023)</i>

- 2022
- Jing Chen, Madhavan Manivannan, Bhavishya Goel, Mustafa Abduljabbar, Miquel Pericàs
 STEER: Asymmetry-aware Energy Efficient Task Scheduler for Cluster-based Multicore Architectures
In IEEE 34th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD 2022)
- Jing Chen, Madhavan Manivannan, Mustafa Abduljabbar, Miquel Pericàs
 ERASE: Energy Efficient Task Mapping and Resource Management for Work Stealing Run-times
In ACM Transactions on Architecture and Code Optimization (TACO 2022)
- 2021
- Jing Chen, Jianbin Fang, Weifeng Liu, Canqun Yang
 BALS: A Blocked Alternating Least Squares Algorithm for Parallel Matrix Factorization
In IEEE Transactions on Parallel and Distributed Systems (TPDS 2021)
- 2020
- Jing Chen, Pirah Noor Soomro, Mustafa Abduljabbar, Madhavan Manivannan, Miquel Pericàs
 Scheduling Task-parallel Applications in Dynamically Asymmetric Environments
In 49th International Conference on Parallel Processing Workshops SRMPDS (ICPPW 2020)
- 2019
- Jing Chen, Madhavan Manivannan, Mustafa Abduljabbar, Miquel Pericàs
 Towards an Energy Aware Task Scheduler for Asymmetric Architectures
In 12th Nordic Workshop on Multi-Core Computing (MCC 2019)
- 2018
- Jing Chen, Jianbin Fang, Weifeng Liu, Tao Tang, Canqun Yang
 clMF: A Fine-Grained and Portable ALternating Least Squares Algorithm for Parallel Matrix Factorization
In Future Generation Computer Systems (FGCS 2018)
- 2017
- Xi Yang, Jianbin Fang, Jing Chen, Chengkun Wu, Tao Tang, Kai Lu
 High Performance Coordinate Descent Matrix Factorization for Recommender Systems
In ACM International Conference on Computing Frontiers (CF 2017)
- Jing Chen, Jianbin Fang, Weifeng Liu, Tao Tang, Xuhao Chen, Canqun Yang
 Efficient and Portable ALS Matrix Factorization for Recommender Systems
In 6th International Conference of Parallel and Distributed Processing Symposium Workshop ParLearning (IPDPSW 2017)
- Jing Chen, Jianbin Fang, Tao Tang, Canqun Yang
 Implementation and Performance Evaluation of Recommender Algorithms Based on Multi-/Many-core Platforms
In Computer Science 2017

Poster Presentations

- 2025 Sonia Rani Gupta, Nikela Papadopoulou, Jing Chen, Miquel Pericàs
CNN and RVV Co-design for Efficient Model Serving
In 19th ACM International Conference on Distributed and Event-Based Systems (DEBS 2025)
- 2019 Jing Chen
An Adaptive Energy & Performance-oriented Task Scheduler for Integrated Systems
In 15th International Summer School on Advanced Computer Architecture and Compilation for High-performance Embedded Systems (ACACES 2019)

Services

- 2026 ACM International Conference on Supercomputing 2026 (ICS 2026)
External Reviewer (2 papers)
- 31st International Workshop on High-level Parallel Programming Models and Supportive Environments (IPDPS HIPS 2026)
Reviewer (2 papers)
- SC 2026 Workshop Proposals
Reviewer (7 proposals)
- 2025 Journal of Systems Architecture
Reviewer
- The SupercomputingAsia and The International Conference on High Performance Computing in Asia-Pacific Region (SCA/HPCAsia 2026)
External Reviewer
- 14th International Workshop on Runtime and Operating Systems for Supercomputers (SC - ROSS)
External Reviewer
- 39th IEEE International Parallel & Distributed Processing Symposium (IPDPS)
External Reviewer
- Design, Automation and Test in Europe Conference (DATE)
External Reviewer
- 2024 The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)
External Reviewer
- ACM International Conference on Supercomputing (ICS)
External Reviewer

2023	ACM International Conference on Supercomputing (ICS) <i>External Reviewer</i>
	The Platform for Advanced Scientific Computing (PASC) <i>Reviewer</i>
2022	International European Conference on Parallel and Distributed Computing (EuroPar) <i>Reviewer</i>
2021	Design, Automation and Test in Europe Conference (DATE) <i>External Reviewer</i>

Supervision

2026 —	Long Cheng, Yicheng Li (Master Students) Supervisor, Chalmers University of Technology (Sweden) Ongoing Master Thesis Project Jonathan Sargent (Master Student) Supervisor, Chalmers University of Technology (Sweden) Ongoing Master Thesis Project Joel Andersson (Ph.D. Student) Co-supervisor, Chalmers University of Technology (Sweden) Research: <i>Efficient usage of the vector register file in RISC-V vector processors for OpenMP and SYCL runtimes</i>
2025 —	Hongguang Chen (Ph.D. Student) Co-supervisor, Chalmers University of Technology (Sweden) Research: <i>High-performance OpenMP and SYCL Tasking Runtime and Compiler Optimization</i>
2025	Long Cheng (Research Assistant) Supervisor, Chalmers University of Technology (Sweden) Research project: <i>KV Cache Quantization Technique in LLMs on GPUs</i> Axel Carlsson, Edvin Mellberg (Master Students) Supervisor, Chalmers University of Technology (Sweden) Master thesis: <i>ILAN: The Interference- and Locality-Aware NUMA Scheduler</i>
2022	Henrik Andersson, Carl Wiede (Master Students) Supervisor, Chalmers University of Technology (Sweden) Master thesis: <i>Energy-Performance Balancing Task Scheduler for Asymmetric Platforms</i>

Teaching

2018 – 2025	High Performance Parallel Programming Chalmers University of Technology (Sweden) <i>Teaching assistant</i>
2021 – 2022	Sustainable Computing Chalmers University of Technology (Sweden) <i>Teaching assistant</i>
2018 – 2021	Parallel Computer Architecture Chalmers University of Technology (Sweden) <i>Teaching assistant</i>

Honors

2015	Outstanding Graduates of Southwest Jiaotong University Southwest Jiaotong University (China)
2014	National Scholarship Southwest Jiaotong University (China)
2013	Excellent Student Leader Southwest Jiaotong University (China)
2012	All-Round Excellence Award Southwest Jiaotong University (China)
2012	Award for Outstanding Spiritual and Ethical Conduct Southwest Jiaotong University (China)
2011 – 2014	Annual Outstanding Undergraduate Student Scholarship Southwest Jiaotong University (China)

Languages

Native	Mandarin Chinese
Fluent	English
Intermediate	Swedish

References

Main Ph.D. supervisor	Prof. Miquel Pericàs Chalmers University of Technology (Sweden) miquelp@chalmers.se
Ph.D. co-supervisors	Dr. Madhavan Manivannan, Dr. Bhavishya Goel Chalmers University of Technology (Sweden) madhavan@chalmers.se , goelb@chalmers.se
Main M.S. supervisor	Prof. Jianbin Fang National University of Defense Technology (China) j.fang@nudt.edu.cn
M.S. co-supervisor	Prof. Weifeng Liu China University of Petroleum-Beijing (China) weifeng.liu@cup.edu.cn